



Introduction to Pharmacodynamics

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Faculty of Medicine

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Objectives

1. **Define** key terms that determine drug Pharmacodynamics including drug receptor agonism and antagonism
2. **Recognize** concepts and terms that are used to quantify drug receptor interactions, including affinity, efficacy and potency
3. **List** types of drug antagonism

Pharmacokinetics

Pharmacodynamics

What happen to the drug?

What the drug do to the body?

- the interactions between the **drug's concentration** and its effect (response)

DOSE



Concentration



Response

(Absorption, Distribution, Metabolism and excretion)

- describe **the action** of the **drug** on the body

NEW TERMS

Antagonist

Affinity

Agonist

Efficacy

Drug Targets :**PROTEINS**

In order of the drug to produce response it has to bind to a certain biological molecule (**target**)

STRUCTURAL

e.g., Tubulin

REGULATORY

Enzyme

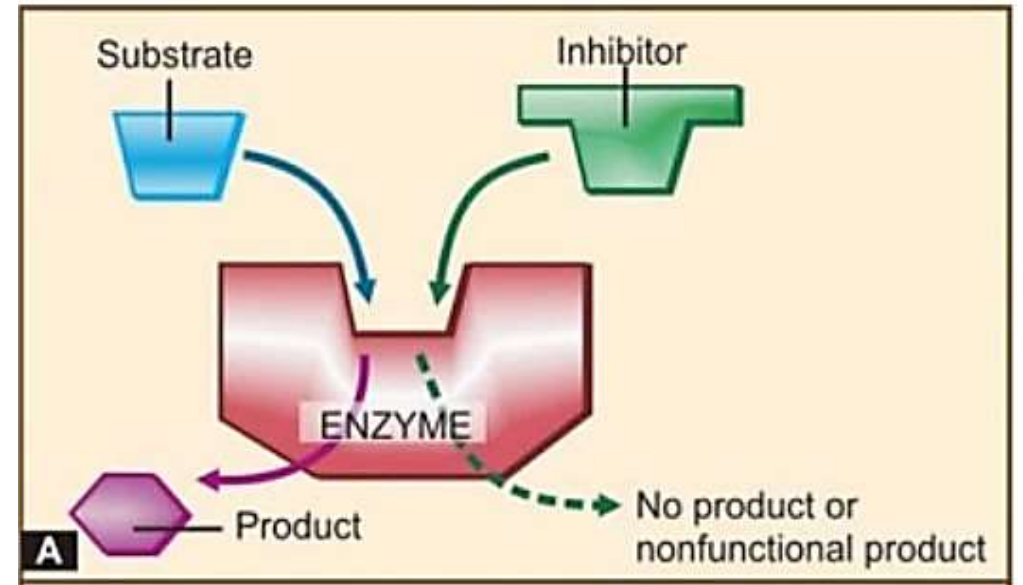
**Carrier
Molecule**

**Ion
Channel**

Receptor

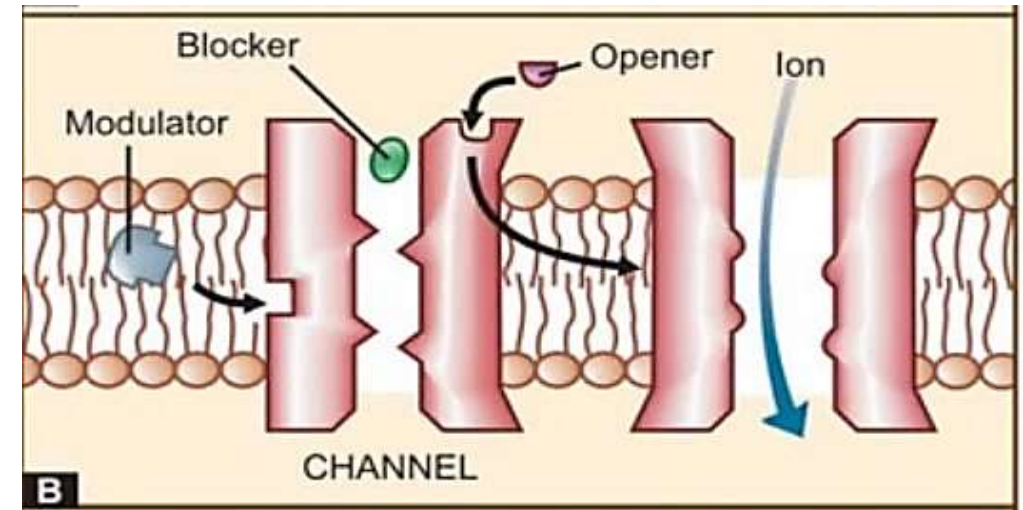
Drug Target: **Enzyme**

Many enzyme are targeted for drugs



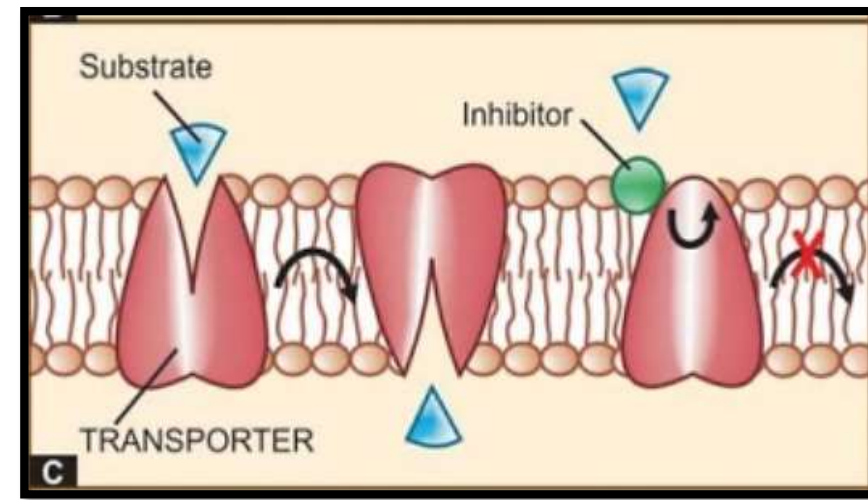
- A. Drug molecule **behave as a substrate** for the enzyme and **competitively inhibit** the enzyme (e.g., neostigmine, is an anti-cholinesterase, compete with acetylcholine for cholinesterase)
- B. Drug may inhibit the enzyme **irreversibly** (e.g., cyclooxygenase enzyme inhibited by **Aspirin**)

Drug Target: **Ion channel**

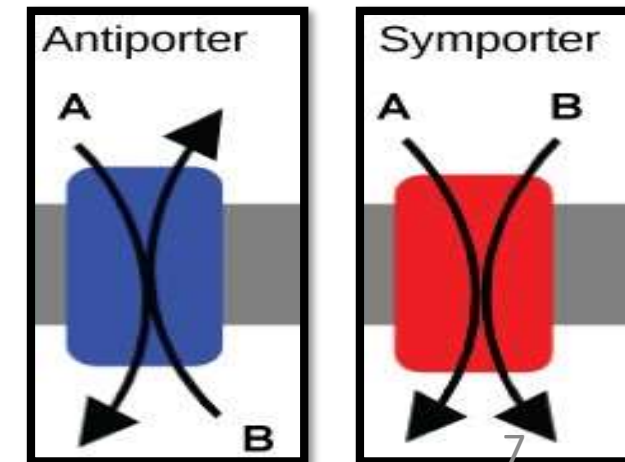


- **Ion channels** are transmembrane protein as pores surrounded by different proteins and regulate intracellular ionic composition.
- **How drugs affect the ion channels ?**
 - A. Drugs can affect ion channels either through **specific receptors (ligand gated ion channels)**
 - B. Ion channels serve as **direct targets** for drug action (e.g. local anesthetics which physically inhibit voltage sensitive Na⁺ channels).
 - C. Certain drugs **modulate** opening and closing of the channels

Drug Target: Carrier Molecule (Transporter)

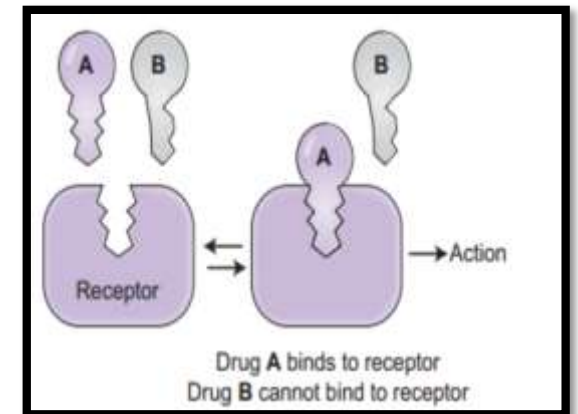
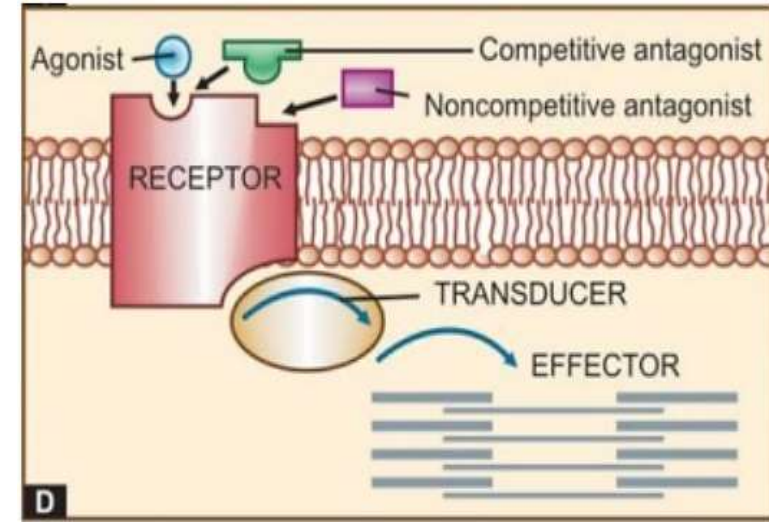


- Small organic molecules and ions are transported across the cell membrane with the aid of certain carrier proteins since these molecules are too polar (i.e. insufficiently lipid soluble to penetrate lipid membranes on their own)
- Examples:**
 - Carriers responsible of the transport of glucose and amino acids into cells
 - Carriers responsible of the transport of sodium and calcium out of cells
 - Carriers responsible of the uptake of neurotransmitter by nerve terminals
- The drug binds to such molecules altering their transport ability.



Drug Target: **Receptors**

- **Receptor** is a macromolecule component present on the cell surface or intracellularly.
- Responsible for selectively sensing & binding of a stimulus (ligand) & its coupling to a response via a set of signal transduction machinery.
- Drugs bind and alter receptor signal transduction machinery.
- Drugs acting on receptors may be:
 - **Agonist**
 - **Antagonist**
 - **Partial agonist or partial antagonist**



Receptors with drugs the same as **lock and key**

Drug Target: **Receptors**

Ligand: (Latin: *ligare*—to bind)

- Any molecule which attaches selectively to particular receptors or sites.
- The term only indicates **affinity** or **binding** without regard to functional change: **agonists** and **competitive antagonists** are both **ligands** of the same receptor.

Agonist

- It initiate changes in cell function & producing effects.
- **It depends on**
 - a. Affinity:** represents **the tendency** of a drug to **bind** to the receptors.
 - b. Efficacy:** the ability of a drug to **produce a desired therapeutic effect**.
- So **agonist** is a drug that has **affinity** & **efficacy**. It interacts with receptors and elicit a response (e.g. adrenaline & acetylcholine)
- Agonist with high efficacy that can produce max effect called **full agonist**.

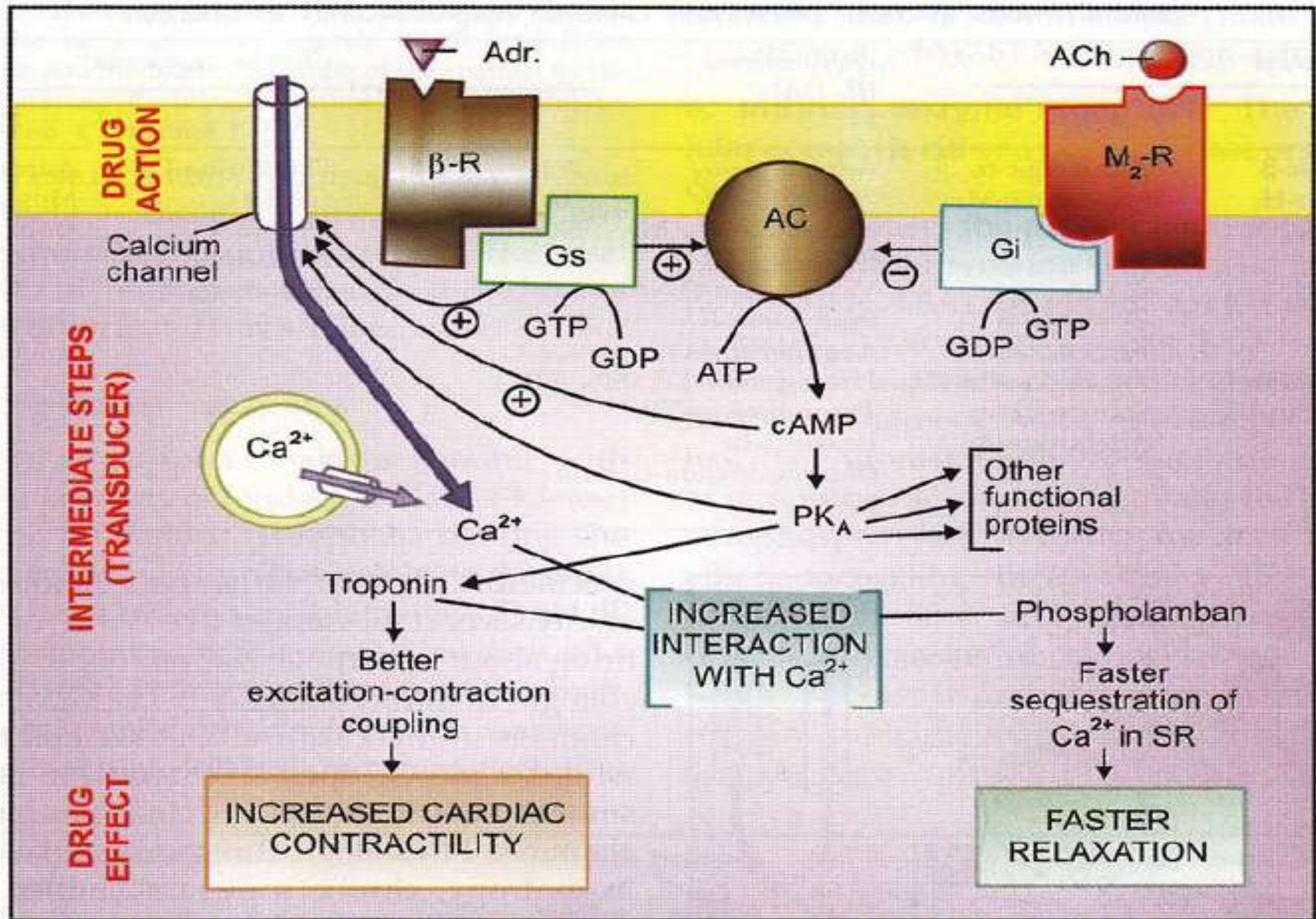
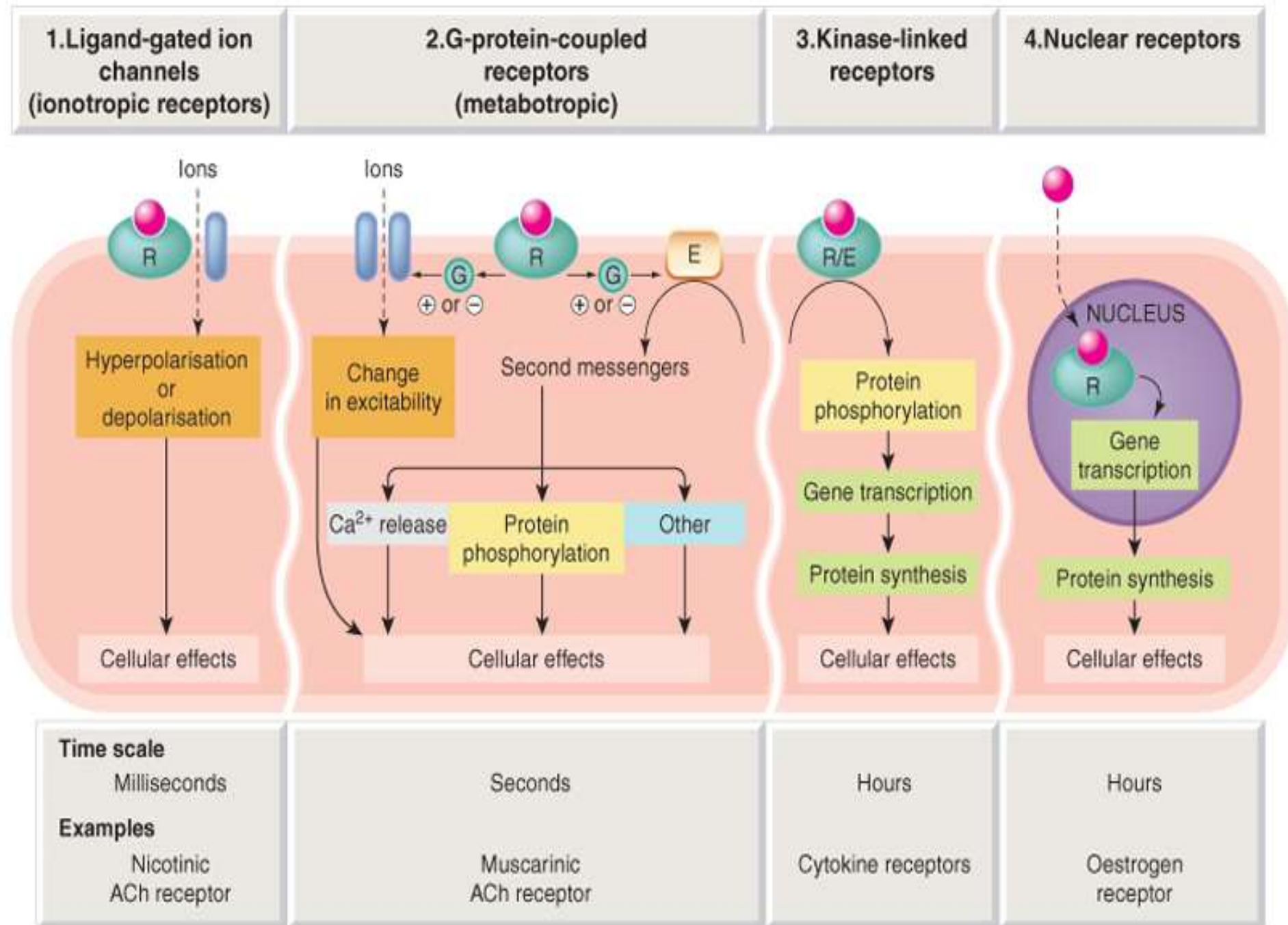


Fig. 4.6: The action-effect sequence of two G-protein coupled (β adrenergic and muscarinic M_2) receptor activation in myocardial cell

Table 4.1: Major functional pathways of G-protein coupled receptor transduction

<i>Adenylyl cyclase: cAMP</i>		<i>Phospholipase IP₃-DAG</i>	<i>Channel regulation</i>		
↑	↓		Ca ²⁺ ↑	Ca ²⁺ ↓	K ⁺ ↑
Adrenergic-β	Adrenergic-α ₂	Adrenergic-α ₁	Adrenergic-β ₁ (Heart, Sk.muscle)	Dopamine-D2	Adrenergic-α ₂
Histamine-H ₂	Muscarinic-M ₂	Histamine-H ₁		GABA _B	Muscarinic-M ₂
Dopamine-D1	Dopamine-D2	Muscarinic-M ₁ , M ₃		Opioid-κ	Dopamine-D2
Glucagon	5-HT ₁	5-HT ₂		Adenosine-A ₁	5-HT _{1A}
FSH & LH	GABA _B	Vasopressin-Oxytocin		Somatostatin	GABA _B
ACTH	Opioid-μ, δ	Bradykinin-B ₂			Opioid-μ, δ
TSH	Angiotensin-AT ₁	Angiotensin-AT ₁			Adenosine-A ₁
Prostaglandin-EP ₂	Prostaglandin-EP ₃	Prostaglandin-FP, EP ₁ , EP ₃			
Prostacyclin-IP	Somatostatin	Thromboxane-TP			
Adenosine-A ₂	Adenosine-A ₁	Leukotriene			
		Cholecystokinin-Gastrin			
		PAF			



Drug Target: **Receptors**

Antagonist

- It binds to the receptor without initiating changes or producing effects
- The **efficacy** is zero
- E.g. Atropine

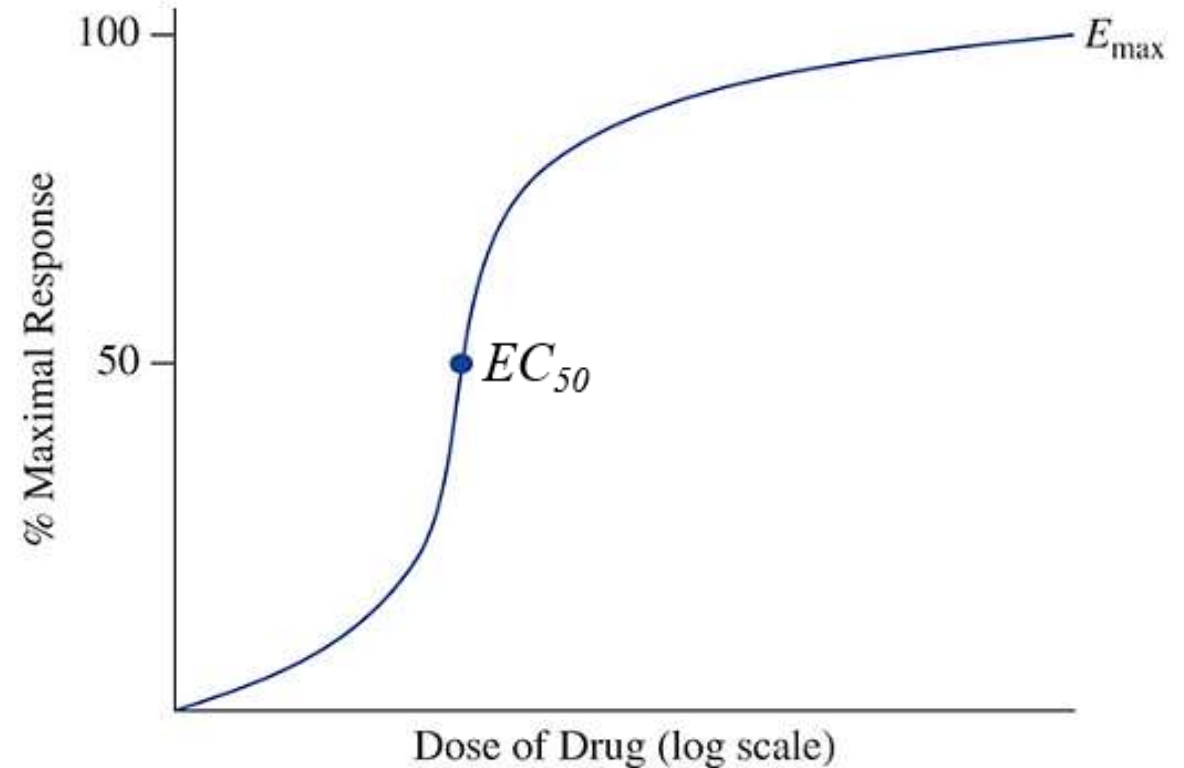
Partial agonist or partial antagonist

- They have **affinity** but some not complete intrinsic activity
- Given alone → they act like an agonist
- Given in combination → they may antagonize the action of a full agonist

Graded dose-response curve

- Charting the change in effect of the drug as the administered drug dose increases.
- Both the potency and efficacy of a drug can be reflected in this graph in terms of EC_{50} and E_{max} respectively.

GRADED DOSE-RESPONSE CURVE



Importance of graded dose-response curves

1. Calculation of the ED50 (The dose that produces 50% of the maximum response in one animal)

- ❖ Comparing ED50 of different drugs on the same animal gives an idea about the **equieffective doses** i.e the doses that produce the same effect.
- ❖ Comparing ED50 of the same drug in different patients gives an idea about the **relative sensitivity** of the patients to the same drug i.e the **degree of biological variation** inherent in a population.

2. Calculation of the maximum response

- ❖ Comparing the maximum response obtained by different drugs on the same organ gives an idea about the **intrinsic activity**.

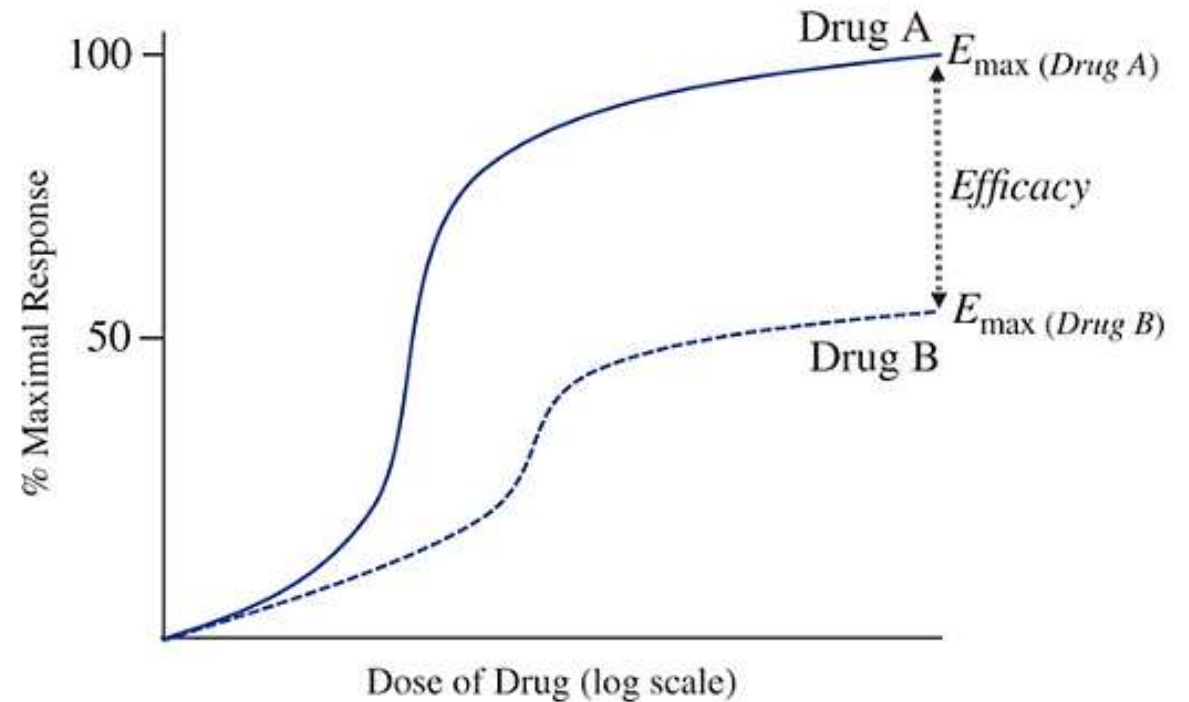
3. Determination of the steepness of the dose response

- ❖ Any **small change** in the drug concentration produces **significant increase** in the tissue response. This means that **maximum response** to the drug could be reached **very fast** and the toxicity could be reached very fast too.

Efficacy

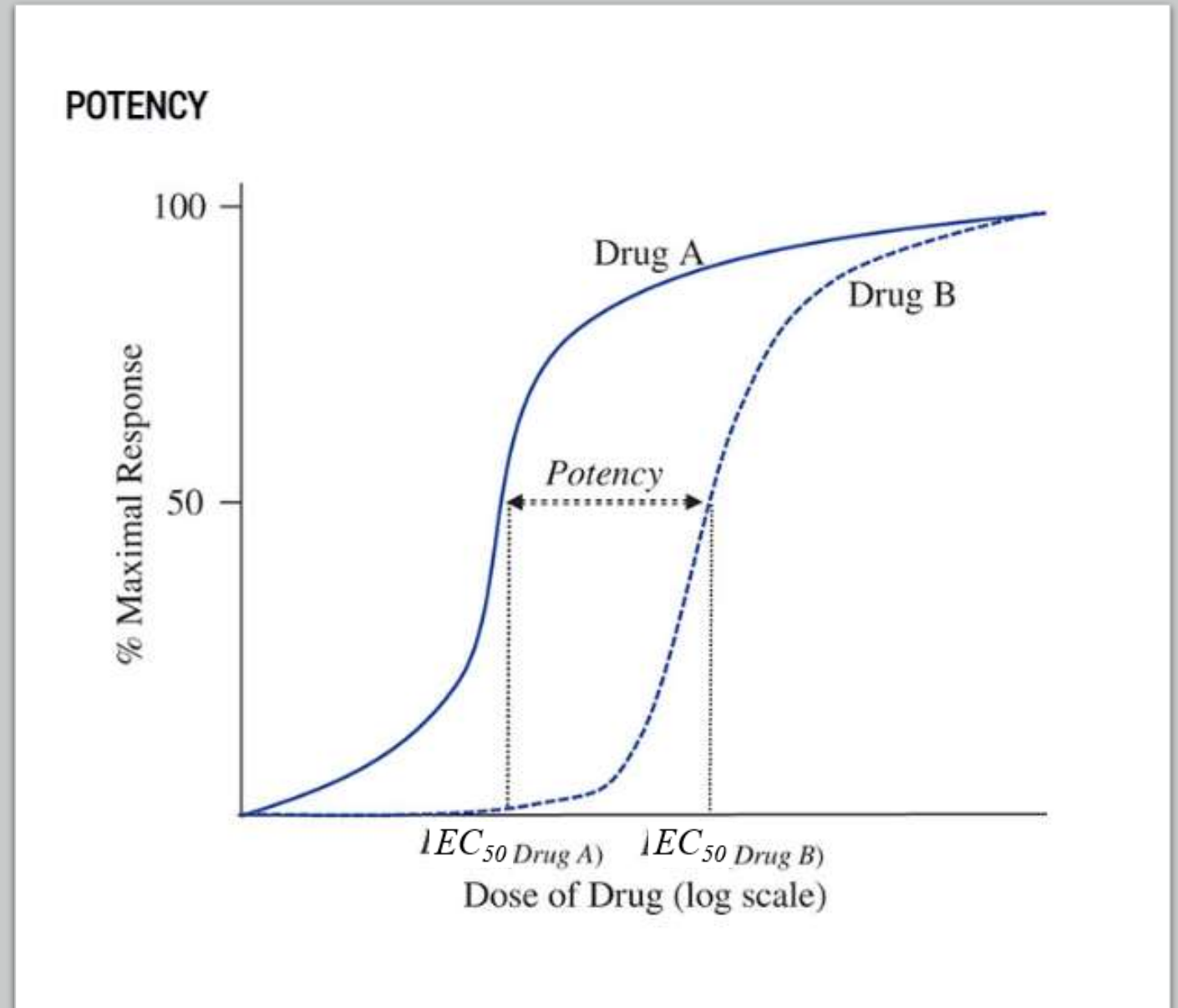
- Refers to the maximal response that can be achieved by a drug ($E_{\max}$).
- As $E_{\max}$ rises, the efficacy of a drug rises as well.
- Efficacy is unrelated to potency (e.g., an efficacious drug can have high or low potency).

EFFICACY



Potency

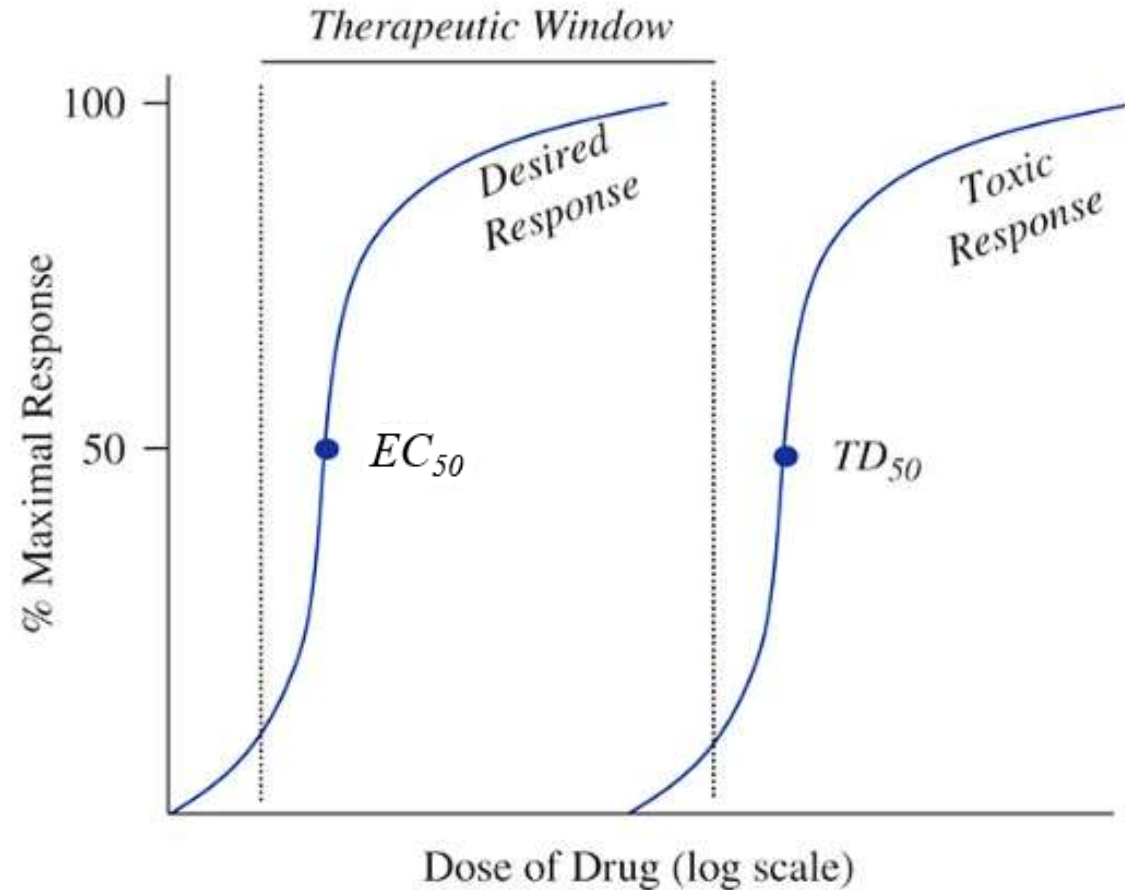
- Refers to the amount of a drug needed to elicit a desired response.
- Drugs of high potency require a small amount of the drug to achieve the desired response.
- The dose of the drug required to achieve 50% of the maximal response is termed EC₅₀
- Potency is unrelated to efficacy (e.g., a potent drug can have high or low efficacy).

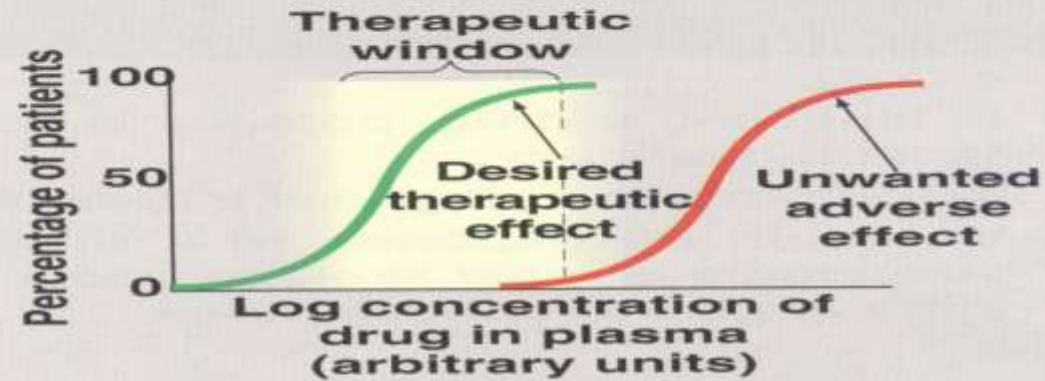
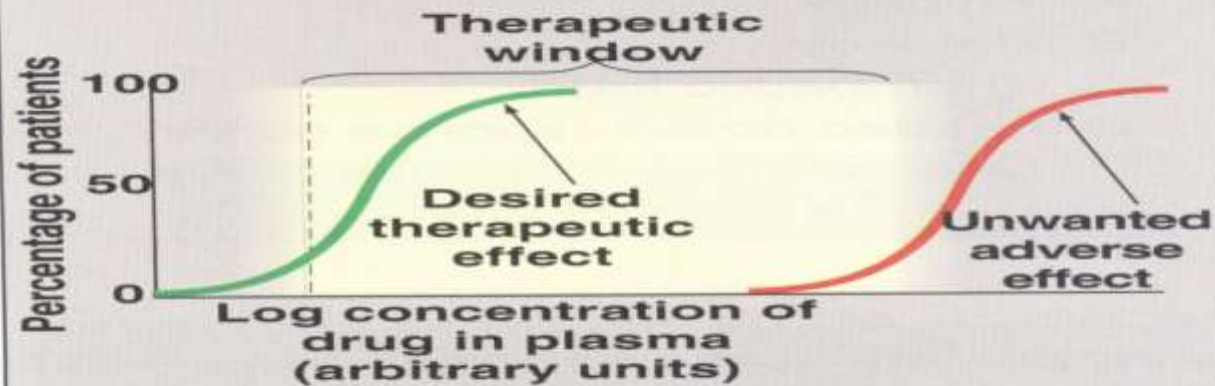


Therapeutic index

- A measure of drug safety
- The therapeutic index compares the median dose of drug necessary to achieve a desired effect (EC_{50}) with the median dose of a drug necessary to produce toxicity (TD_{50}).
- The **therapeutic window** refers to the doses of drugs that can be given, which will produce the desired effect without producing unwanted toxic side effects.

DETERMINING THE THERAPEUTIC INDEX



A**Warfarin: Small therapeutic index****B****Penicillin: Large therapeutic index****Figure 2.11**

Cumulative percentage of patients responding to plasma levels of a drug.

Antagonism

It is the decrease or the complete abolishment of the effect of one drug in the presence of another.

Types

- **Physiological antagonism (Functional antagonism)**
- **Chemical antagonism**
- **Pharmacokinetic antagonism**
- **Pharmacological antagonism**
 - **Competitive:** Reversible or Irreversible
 - **Non-competitive**

Antagonism

- **Physiological antagonism (Functional antagonism):** two drugs act on different receptors to produce different physiological effects. e.g. Histamine (↓BP) & Adrenaline (↑BP)
- **Chemical antagonism** (Reaction between 2 compounds → loss of activity)
- **Pharmacokinetic antagonism** (antagonist effectively reduces conc of the active drug at its site of action)
- **Pharmacological antagonism**
 1. **Competitive:** (Reversible or Irreversible)
 2. **Non-competitive**

Competitive Vs Non-Competitive antagonist

Competitive (equilibrium type)

1. Antagonist binds with the same receptor as the agonist
2. Antagonist resembles chemically with the agonist
3. Parallel rightward shift of agonist DRC
4. The same maximal response can be attained by increasing dose of agonist (surmountable antagonism)
5. The antagonist appears to have inactivated a certain number of agonist molecules
6. Intensity of response depends on the concentration of both agonist and antagonist
7. Examples: ACh—Atropine
Morphine—Naloxone

Noncompetitive

- Binds to another site of receptor
- Does not resemble
- Flattening of agonist DRC
- Maximal response is suppressed (unsurmountable antagonism)
- The antagonist appears to have inactivated a certain number of receptors
- Maximal response depends only on the concentration of antagonist
- Diazepam—Bicuculline

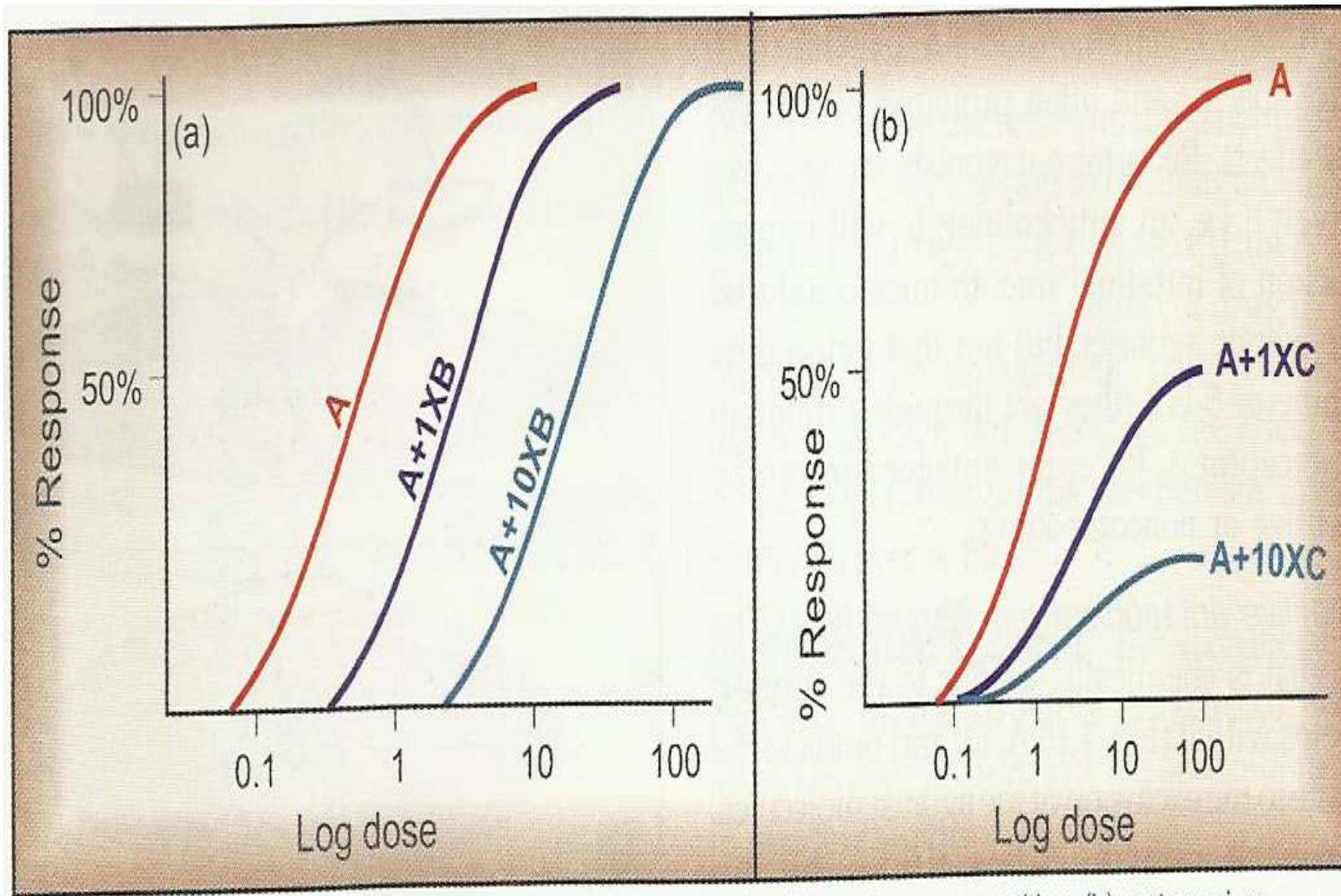
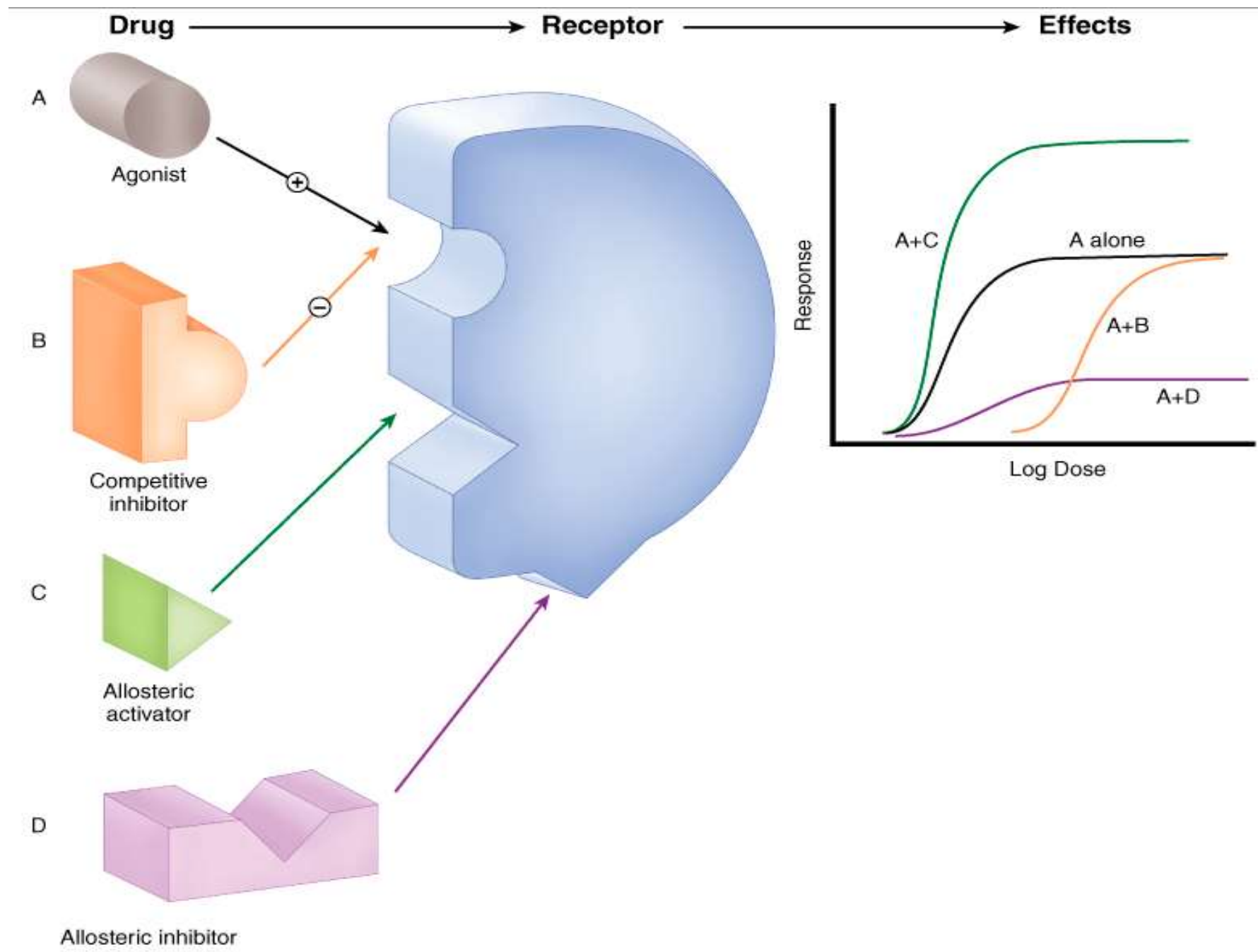


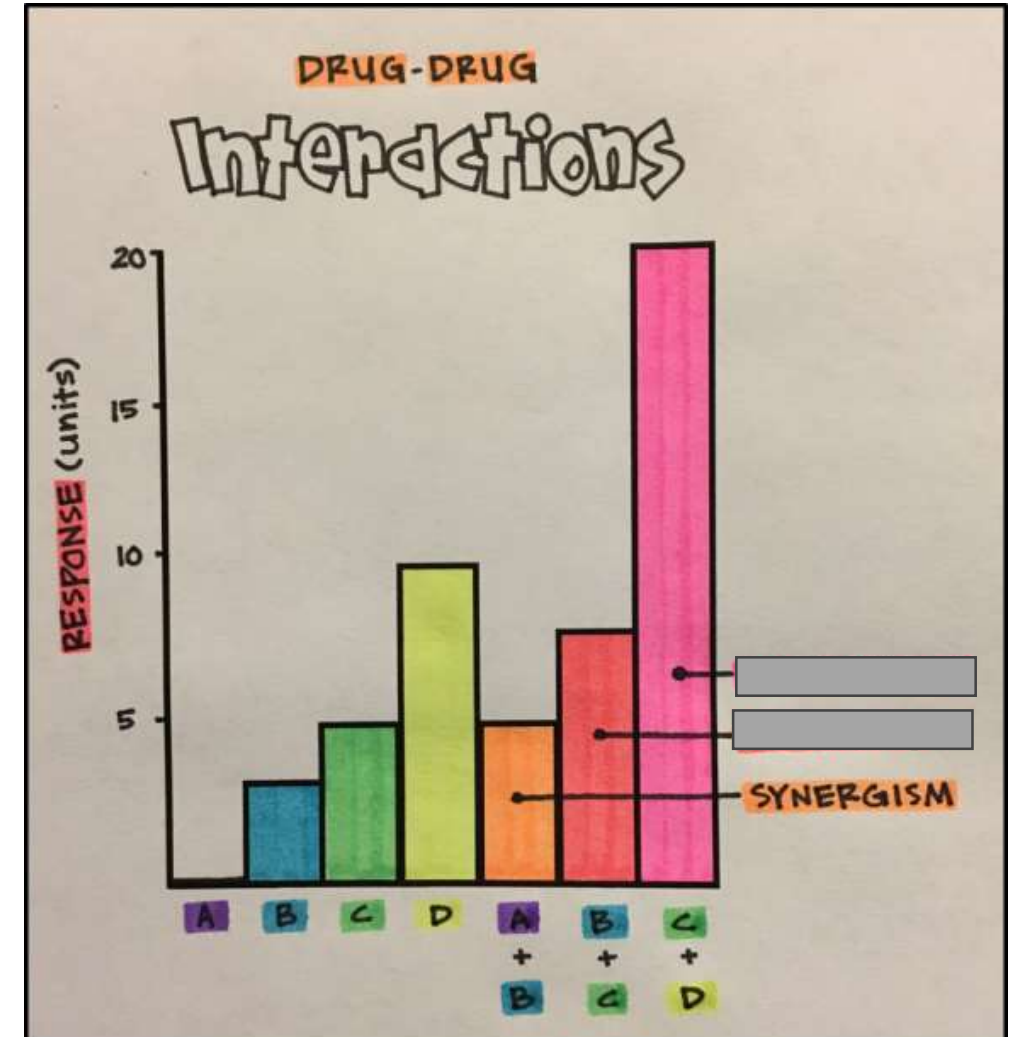
Fig. 4.17: Dose-response curves showing competitive (a) and noncompetitive (b) antagonism
A—agonist, B—competitive antagonist, C—noncompetitive antagonist



SYNERGISM

(Greek: **Syn**-together; **ergon**-work)

- When the **action of one drug is facilitated or increased by the other**, they are said to be **synergistic**.
- In a synergistic pair, both the drugs can have action in the same direction or given alone one may be inactive (**Drug A**) but still enhance the action of the other (**Drug B**) when given together.

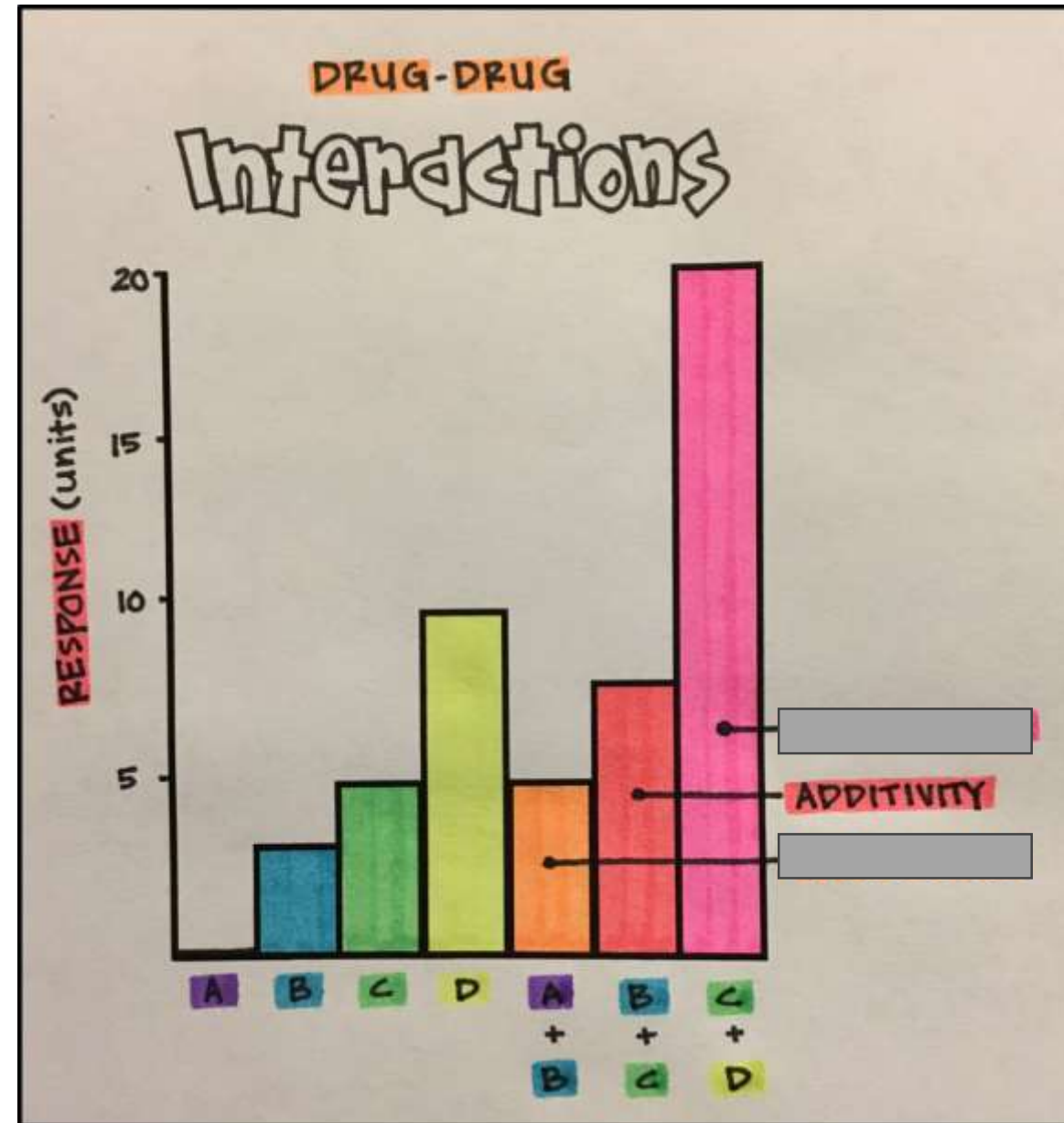


Additive

The effect of the two drugs is in the same direction and simply adds up:
effect of drugs B + C = effect of drug B + effect of drug C

Additive drug combinations

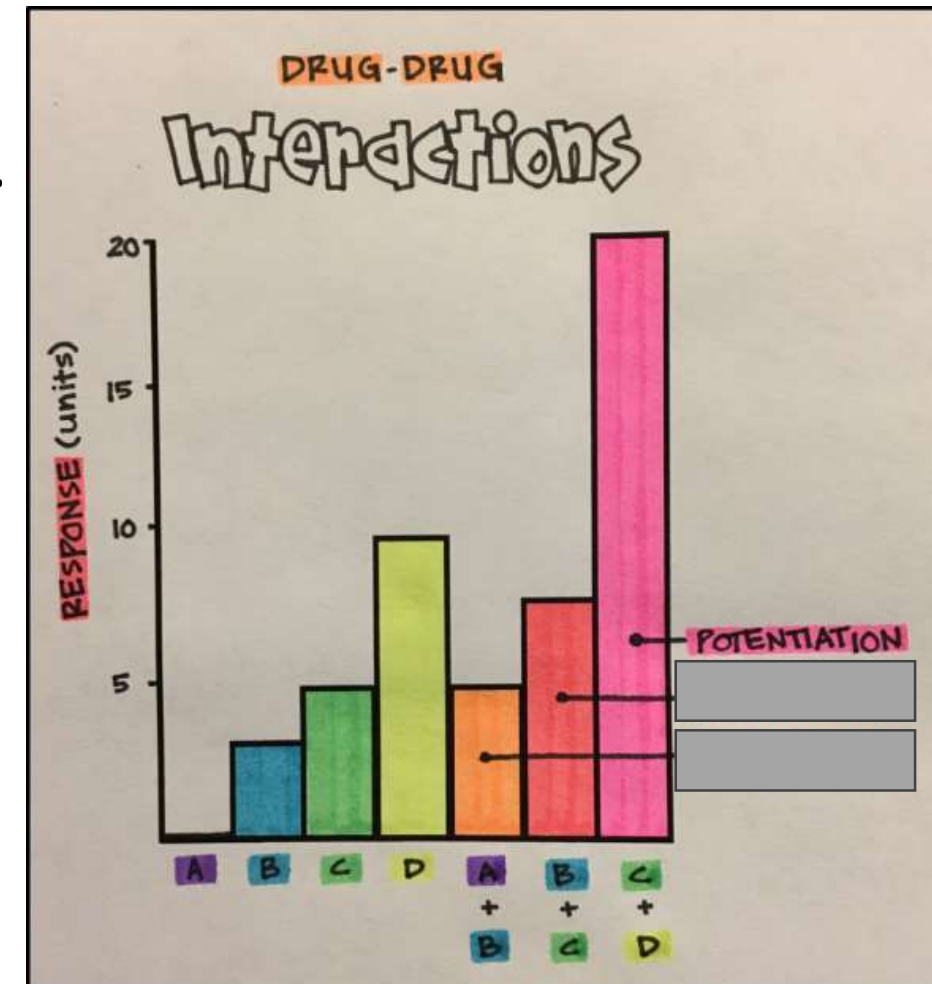
Aspirin + paracetamol	as analgesic/ antipyretic
Nitrous oxide + halothane	as general anaesthetic
Amlodipine + atenolol	as antihypertensive
Glibenclamide + metformin	as hypoglycaemic
Ephedrine + theophylline	as bronchodilator



Supraadditive (potentiation)

- The effect of combination is greater than the individual effects of the components:
 effect of drug C + D > effect of drug C + effect of drug D
 This is always the case when one component is inactive as such.

Supraadditive drug combinations	
Drug pair	Basis of potentiation
Acetylcholine + physostigmine	Inhibition of break down
Levodopa + carbidopa/benserazide	Inhibition of peripheral metabolism
Adrenaline + cocaine/desipramine	Inhibition of neuronal uptake
Sulfamethoxazole + trimethoprim	Sequential blockade
Antihypertensives (enalapril + hydrochlorothiazide)	Tackling two contributory factors
Tyramine + MAO inhibitors	Increasing release-able CA store



Undesirable effects of drugs

Tolerance

- It is described as the failure of responsiveness to the usual dose of drug.
- This means the drug dosage must be increased to maintain a given therapeutic effect.

Tachyphylaxis (special type of acquired tolerance)

- It is a type of tolerance which occur very rapidly (few minutes) → called Acute acquired tolerance.

Feedback is anonymous and highly appreciated

